

1 Review of limits

Question 1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function and $x_0 \in \mathbb{R}$ be a point. State the definition of $\lim_{x \rightarrow x_0} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.

Question 2. Find the following limits or determine that they do not exist:

a) $\lim_{x \rightarrow \infty} \left(5 + \frac{1}{x}\right)$

b) $\lim_{x \rightarrow \infty} \frac{\sqrt{5x^2 - 3}}{3x + 2}$

c) $\lim_{x \rightarrow \pi/2} \frac{11(\tan x)^2 + x}{2(\tan x)^2 + 1}$

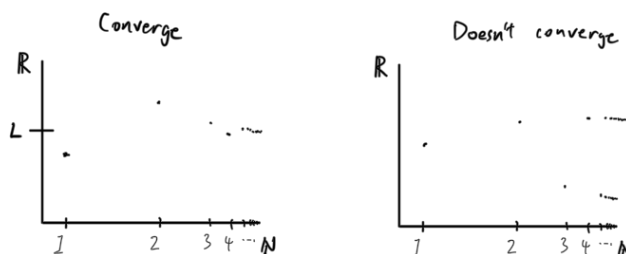
d) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

e) $\lim_{x \rightarrow \infty} x \sin(1/x)$

f) $\lim_{x \rightarrow \infty} \frac{e^{x^2}}{xe^x}$

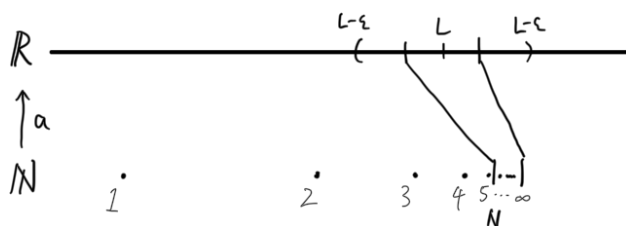
2 Sequential limits

A sequence a_n converges if as n gets large, a_n gets closer and closer to a real number L .



The formal definition of sequential limits is as follows:

$\lim_{n \rightarrow \infty} a_n = L$ if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|a_n - L| < \epsilon$ for all $n \geq N$.



The tools we developed for limits of functions can be used to find limits of sequences thanks to the theorem:

If $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} a_n = L$ where $a_n = f(n)$.

Question 3. Which of the following sequences converge? If they do, find the limit.

a) $a_n = \frac{3n+2}{n^2+1}$

b) $a_n = \frac{(n+1)! \cdot n^2}{(n+3)!}$

c) $a_n = (-1)^n$

d) $a_n = \cos(2\pi n)$

e) $a_1 = 1$ and $a_{n+1} = a_n + \left(\frac{1}{n+1} - \frac{1}{n}\right)$

f) $a_n = \frac{4^n}{n}$

g) $a_n = \frac{\ln n}{n}$

h) $a_n = \frac{\ln n}{n^{0.01}}$

i) $a_n = (n+1) \sin(1/n)$

j) $a_n = n \ln\left(1 + \frac{1}{n}\right)$

k) $a_1 = 1$ and $a_{n+1} = \frac{5+a_n}{2}$

l) $a_n = \frac{n^{100}}{2^n}$

3 Continuous function theorem

Question 4. Argue from definitions that if $\lim_{n \rightarrow \infty} a_n = L$ and if f is a function that is continuous at L (i.e., $\lim_{x \rightarrow L} f(x) = f(L)$), then $\lim_{n \rightarrow \infty} f(a_n) = f(L)$.

Question 5. Find the limits of the following sequences or determine that they do not exist:

a) $\lim_{n \rightarrow \infty} \frac{\sqrt{n+1}}{\sqrt{n+3}}$

b) $\lim_{n \rightarrow \infty} (\ln n - \ln(n+1))$

c) $\lim_{n \rightarrow \infty} \sqrt[n]{n^2+1}$

d) $\lim_{n \rightarrow \infty} \left(\frac{n}{n+x}\right)^n$ (any positive real x)

4 Sandwich theorem

Sandwich theorem states that if $a_n \leq b_n \leq c_n$ for all $n \geq n_0$, then (assuming the limits exist)

$$\lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n \leq \lim_{n \rightarrow \infty} c_n.$$

Question 6. Find the limits of the following sequences or determine that they do not exist:

a) $\lim_{n \rightarrow \infty} \frac{\cos(n)}{n}$

b) $\lim_{n \rightarrow \infty} \frac{(-1)^n \cdot n}{1+n}$

c) $\lim_{n \rightarrow \infty} \frac{3^n}{n!}$

d) $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$ (hint: compare with $1/n$)

e) $\lim_{n \rightarrow \infty} \frac{2^{(n+1)n}}{n!}$ (hint: $2^{(n+1)n} = 4^n 4^{n-1} \dots 4^2 4^1$)